

# **Engineering a Compiler**

## **2nd Edition**

### **Chapter 7**

### **Code Shape**

Figure 7.1

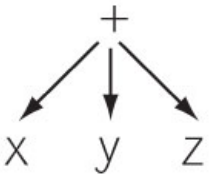
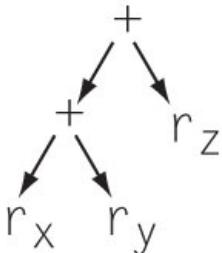
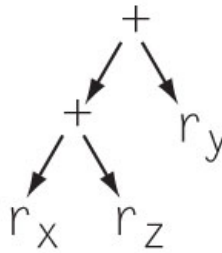
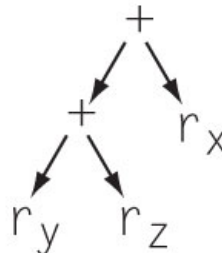
| Source Code |                                                                                   | Low-Level, Three-Address Code                                                      |                                                                                     |                                                                                     |
|-------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Code</b> | $x + y + z$                                                                       | $r_1 \leftarrow r_x + r_y$<br>$r_2 \leftarrow r_1 + r_z$                           | $r_1 \leftarrow r_x + r_z$<br>$r_2 \leftarrow r_1 + r_y$                            | $r_1 \leftarrow r_y + r_z$<br>$r_2 \leftarrow r_1 + r_x$                            |
| <b>Tree</b> |  |  |  |  |

Figure 7.2

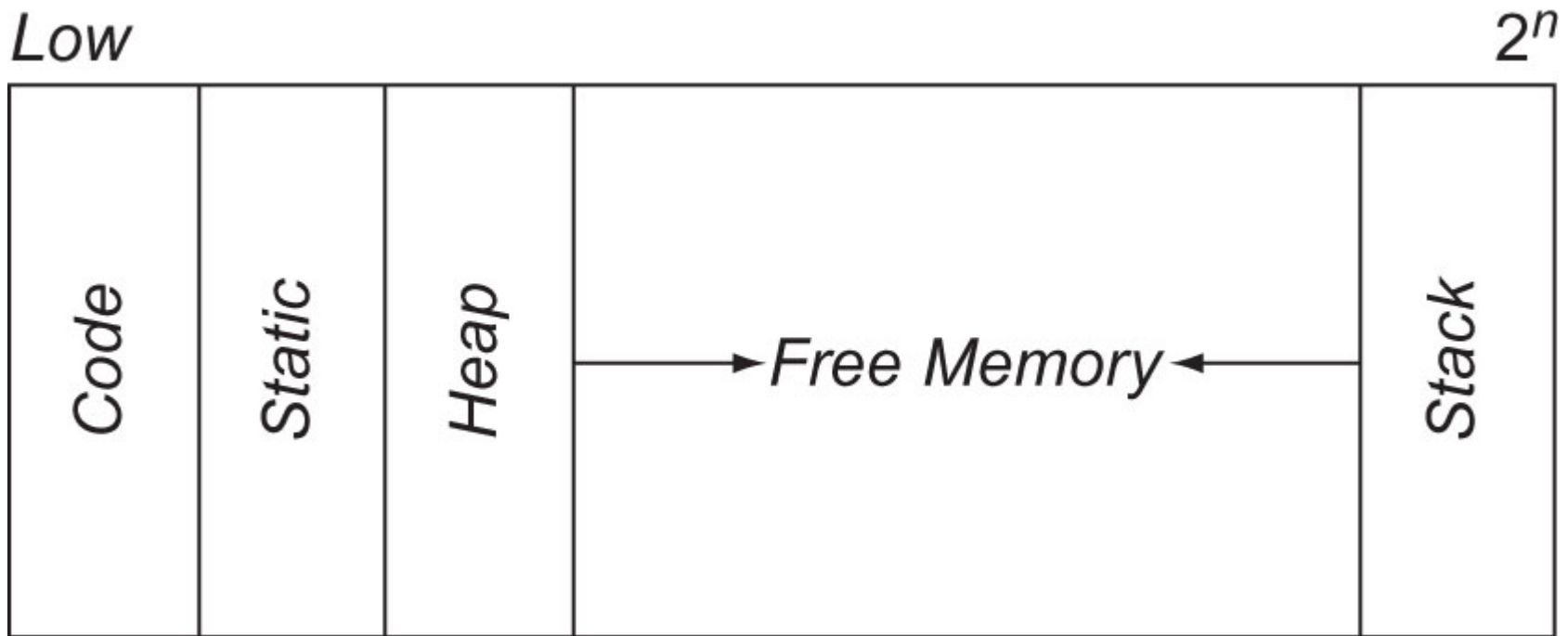


Figure 7.3

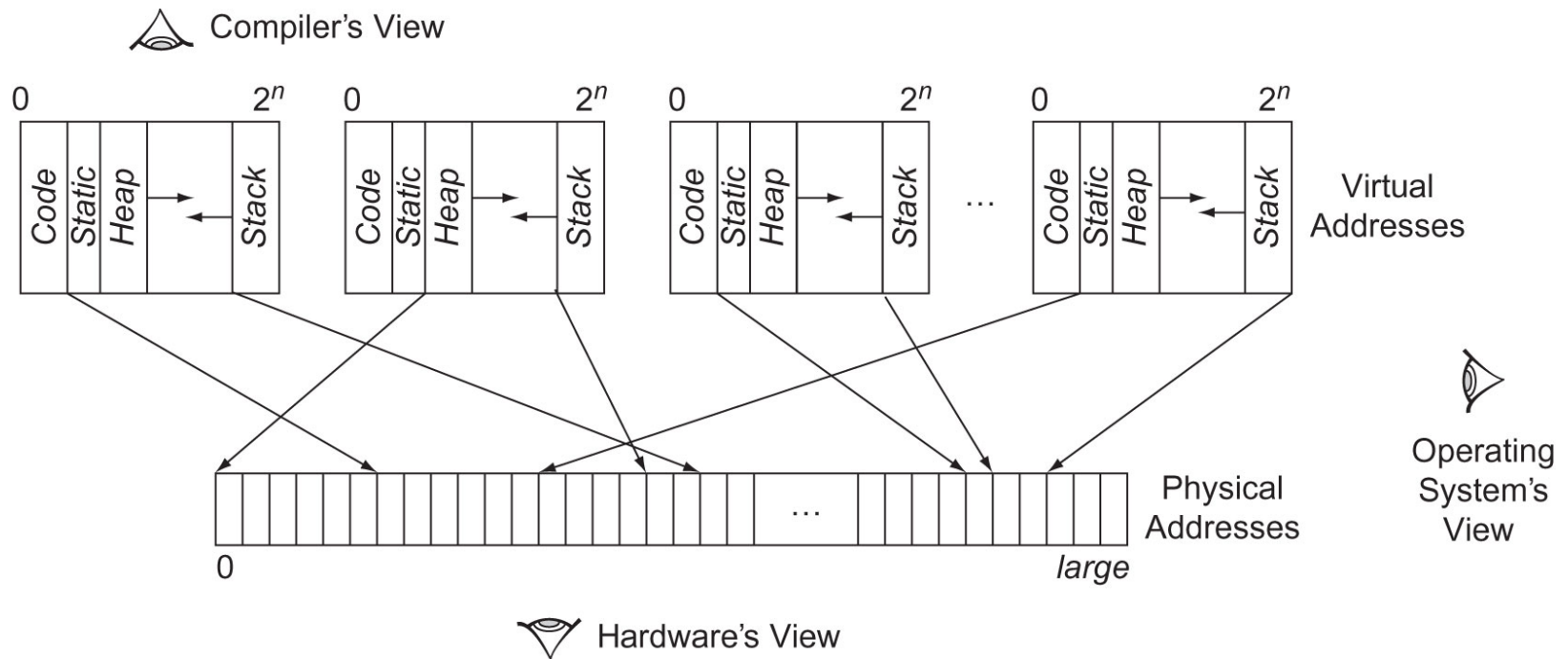
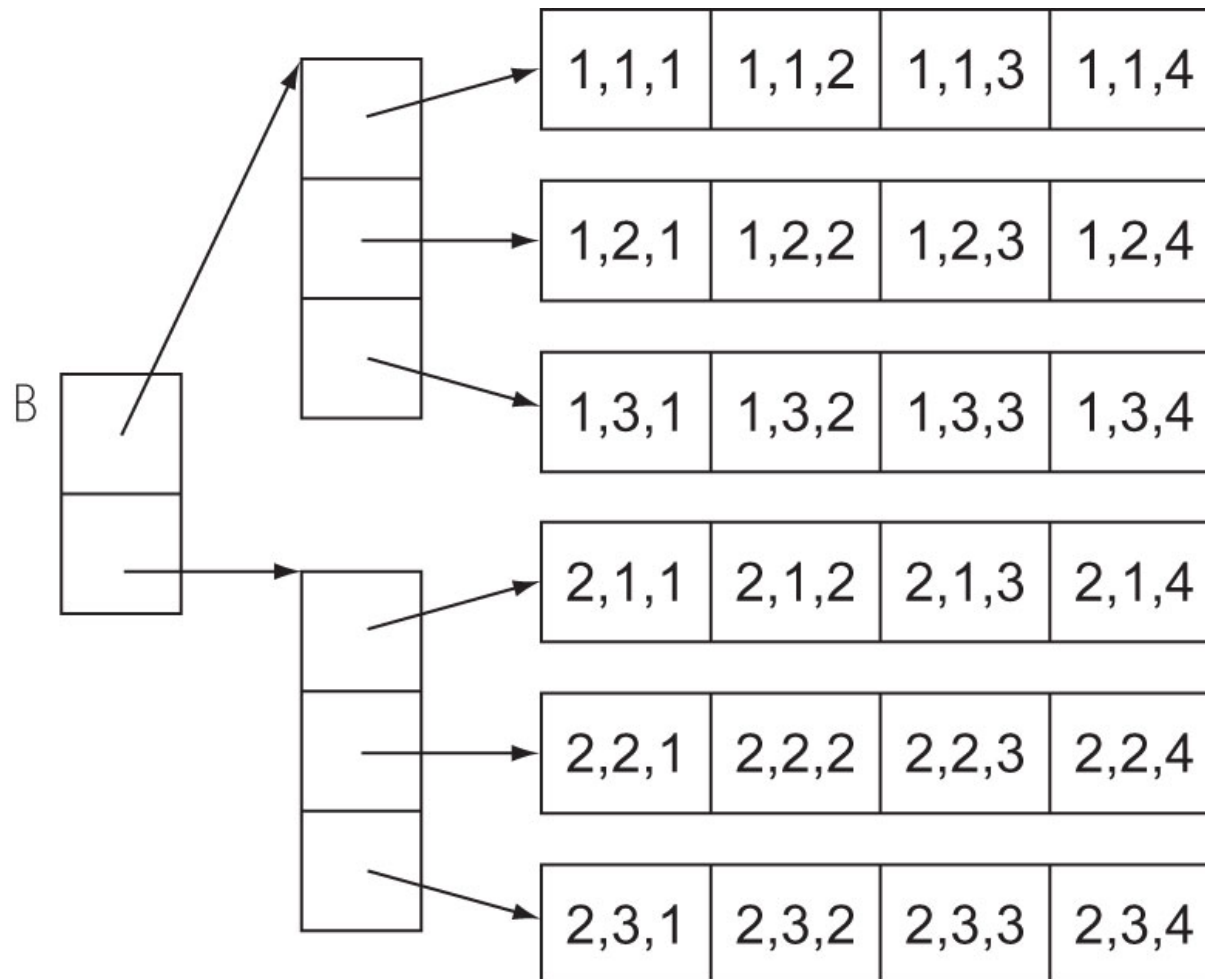
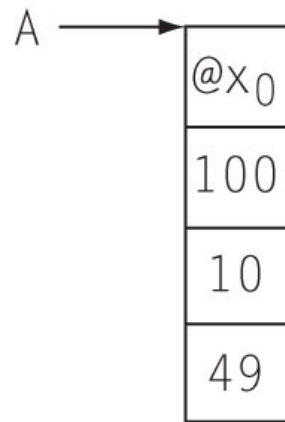
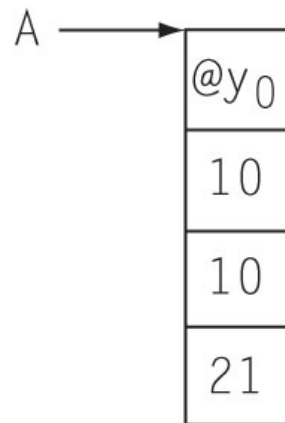


Figure 7.10



**Figure 7.11**

At the First Call



At the Second Call

Figure 7.12

Structure Layout Table

| Name | Length | 1 <sup>st</sup> Element |
|------|--------|-------------------------|
| node | 8      | •                       |
| ...  | ...    | •                       |

Structure Element Table

| Name       | Length | Offset | Type          | Next |
|------------|--------|--------|---------------|------|
| node.value | 4      | 0      | int           | •    |
| node.next  | 4      | 4      | struct node * | •    |
| ...        | ...    | ...    | ...           | ...  |

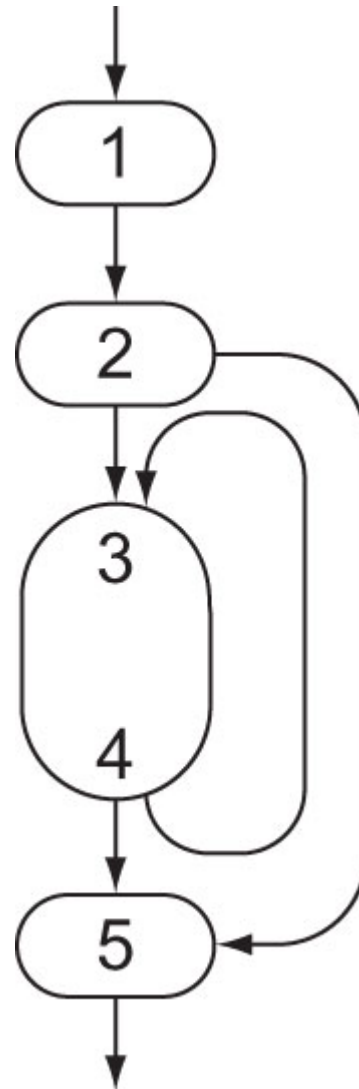
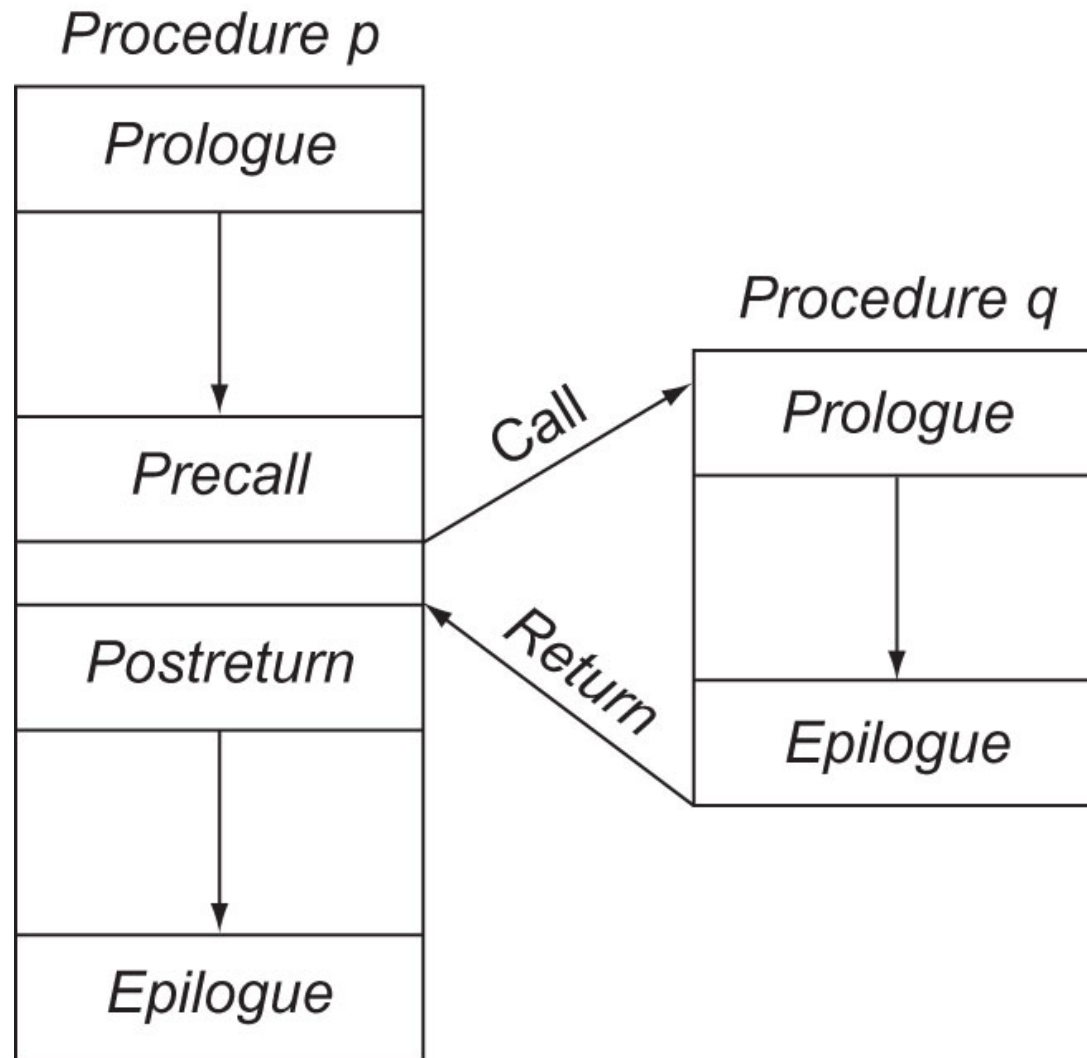
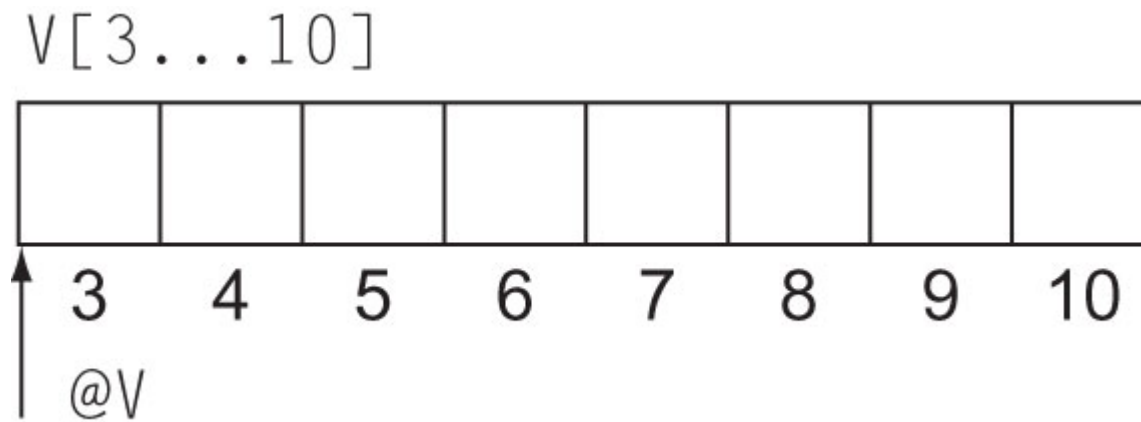
**Figure 7.15**



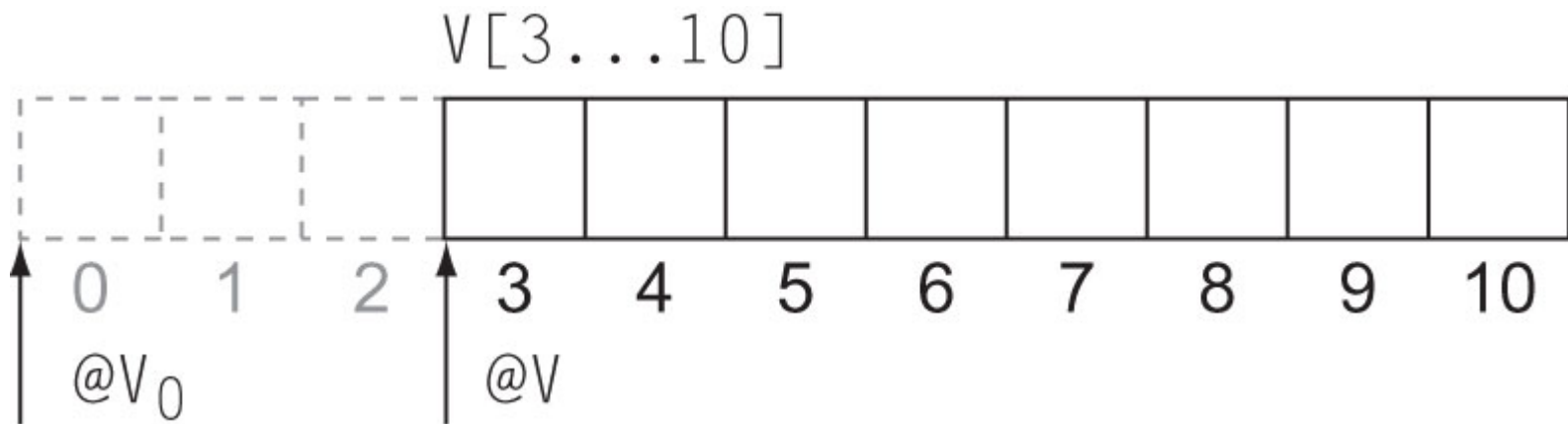
Figure 7.19



## Un - Figure 7.1



## Un - Figure 7.2



## Un - Figure 7.3

A

|     |     |     |     |
|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 |
| 2,1 | 2,2 | 2,3 | 2,4 |

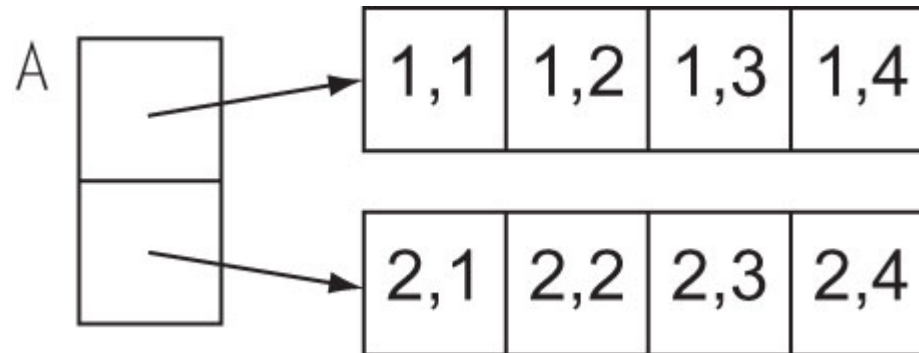
## Un - Figure 7.4

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 2,1 | 2,2 | 2,3 | 2,4 |
|-----|-----|-----|-----|-----|-----|-----|-----|

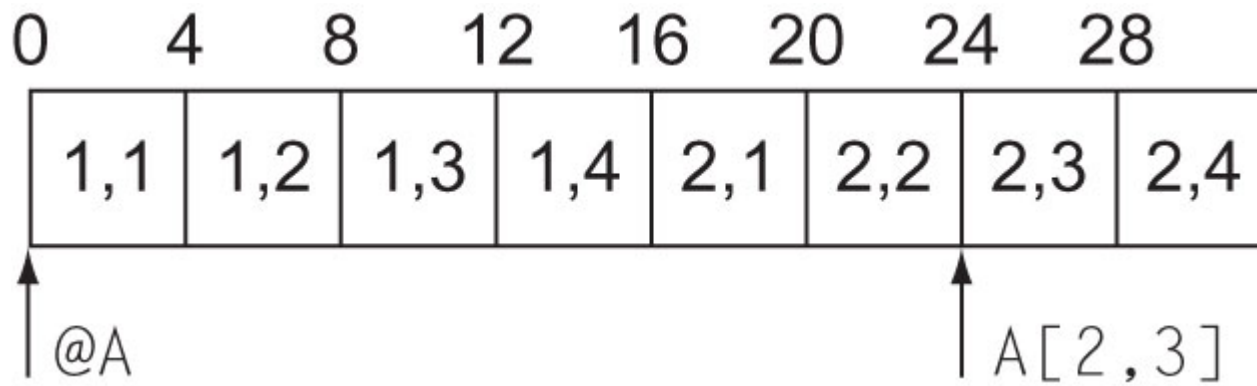
## Un - Figure 7.5

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| 1,1 | 2,1 | 1,2 | 2,2 | 1,3 | 2,3 | 1,4 | 2,4 |
|-----|-----|-----|-----|-----|-----|-----|-----|

## Un - Figure 7.6

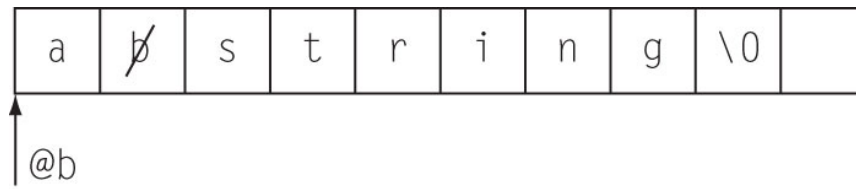


## Un - Figure 7.7

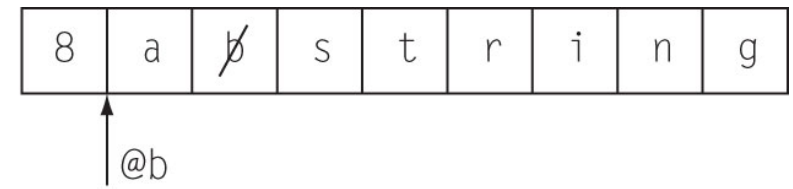




## Un - Figure 7.8

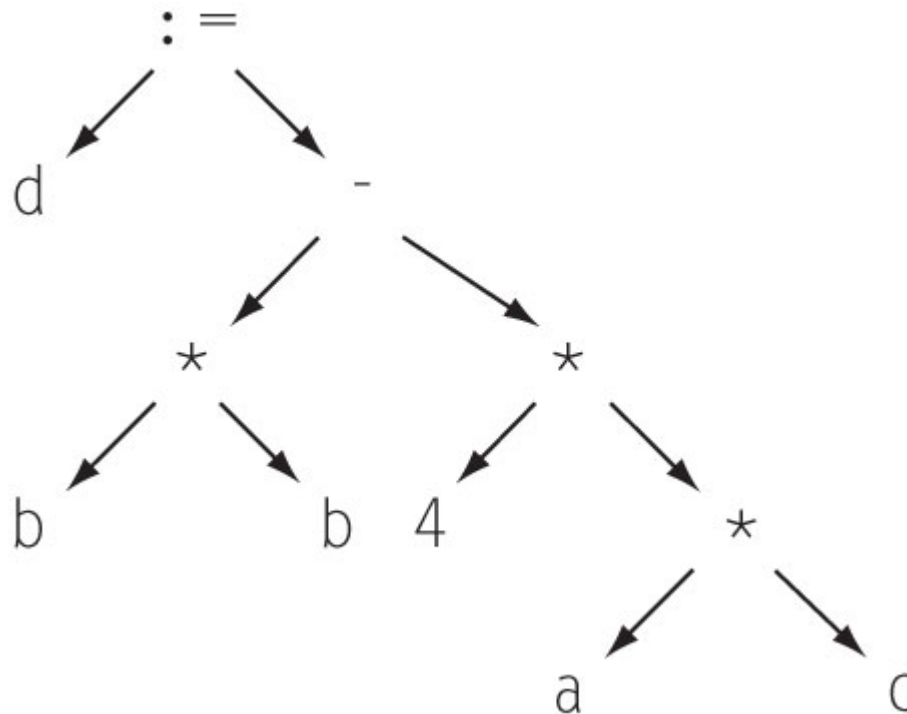


Null Termination



Explicit Length Field

## Un - Figure 7.9



## Un - Figure 7.10

